

Mian Saad Karim

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PROFILE

Hardware and Embedded Systems Engineer with hands-on experience in PCB design, custom flight-controller firmware, edge AI deployment, and UAV swarm research. Proficient across the full product lifecycle—from schematic capture and 4-layer PCB layout to firmware bring-up (ESP-IDF, ArduPilot) and field-tested drone assembly.

EDUCATION

Pakistan Institute of Engineering & Applied Sciences (PIEAS) Graduated Jul 2026
BS in Electrical Engineering CGPA: 3.64

EXPERIENCE

Hardware Engineer, *Signatics* Jul 2026 – Present

- Designing custom high-power multi-rail power distribution systems to automate testing and verification of various products.

Electrical Engineer (Part-time), *Saf Shikan* Sep 2025 – Jun 2026

- Led development of mini-drones based on ESP32 microcontrollers.
- Wrote and deployed custom flight-controller firmware for a SWARM UAV research project.

Intern, *Saf Shikan* Jul 2025 – Sep 2025

- Assembled, configured, and calibrated Pixhawk-based drones; conducted autonomous flights via Raspberry Pi.
- Designed and prototyped a custom mini-drone from scratch (ESP32 + IMU + optical-flow sensor).

Intern – Quest Internship Program, *NASTP* Jul 2025 – Aug 2025

- Contributed to prototyping projects and STEM educational kits; secured NASTP budget funding.

Intern, *Pakistan Aeronautical Complex (PAC) Kamra* Jul 2024

- Gained hands-on experience in avionics production, RF testing, and aircraft manufacturing operations.

PROJECTS

FYP: Custom UAV Platform for Autonomous Disaster Assessment [GitHub](#)

- Engineered a dual-architecture **4-layer PCB**: STM32F405 (ArduPilot) + ESP32-S3 (edge AI, telemetry, swarm comms).
- Deployed onboard fire-detection AI via Edge Impulse; ESP-IDF firmware enables autonomous MAVLink mission execution.
- PCB doubles as structural frame—reducing weight and eliminating loose wiring.

ESP32-S3 Fire Detection + Pixhawk Companion Computer [GitHub](#)

- Combined Edge Impulse FOMO fire detection with a MAVLink v2 companion computer on an ESP32-S3.
- Web dashboard: live camera feed with AI overlay, arm/disarm, RC override, auto-flight, and full telemetry.

Flying Mini Clone: 3-PCB Modular Flight Stack Fiverr Freelance

- Ground-up PCB clone of SpeedyBee F405 Wing Mini: FC (STM32F405, ICM-42688-P, AT7456E OSD), PDB, USB-C breakout.

FPGA-Based Dual-Axis Servo Control with UART + GUI [GitHub](#)

- Real-time servo control in Verilog on Spartan-3E FPGA with Python/Tkinter GUI over UART.

VGA Graphics & Game on Spartan-3E FPGA [GitHub](#)

- Minimalist airplane game in Verilog generating 640×480 VGA signal entirely through RTL.

TECHNICAL SKILLS

PCB Design: Altium Designer, EasyEDA (schematic + layout), Gerber generation, JLCPCB; 2-layer and 4-layer stackups

MCUs & Firmware: ESP32, ESP32-S3, STM32F405; ESP-IDF, ArduPilot, INAV

Protocols: MAVLink v2, ESP-NOW, SPI, I²C, UART, USB-C

FPGA / HDL: Verilog on Xilinx Spartan-3E (VGA, UART, PWM generation)

Edge AI: Edge Impulse (FOMO object detection), TinyML on ESP32-S3
Languages & Tools: C, C++, Python, Verilog, MATLAB, Git/GitHub, Linux CLI, Onshape CAD

CERTIFICATIONS

FPV Drone Workshop – Saf Shikan	Jul 2025
Intro to Parametric Feature-Based CAD – Onshape/PTC	Jan 2025
Rocket Science for Everyone – Yale University	Jan 2025
ISYWSC’24 Runner-up, Electro Masters – IEEE	Dec 2024
Career Essentials in GitHub – GitHub	Sep 2024
Learning FPGA Development – LinkedIn Learning	Aug 2024
CS50x – Harvard / CS50	Jul 2024